

Simple ICMP File Transfer Specification

Packet Layout

Each data packet uses only the ICMP identifier and sequence fields. Option A: - ICMP identifier = 2 bytes of file data. - ICMP sequence = packet sequence number. Chunk size: - Use sequence values 0-1023. - Maximum 1024 packets per chunk. - 2 bytes per packet = 2048 bytes per chunk.

Metadata Packets

Packet 1: Filename length. Packet 2: File size part 1 (upper 16 bits). Packet 3: File size part 2 (lower 16 bits). Packet 4 onward: Filename bytes. After filename packets come the file data packets.

Sender Flow

1. Read file into bytes. 2. Send packet 1 containing filename length. 3. Send packet 2 and 3 containing file size. 4. Send filename bytes. 5. Split file into chunks. 6. Send chunk 0. 7. Wait for ACK. 8. Send chunk 1. 9. Repeat until complete.

Receiver Flow

1. Read packet 1 and obtain filename length. 2. Read packet 2 and 3 and reconstruct file size. 3. Receive filename bytes. 4. Start receiving chunk 0. 5. Store packets by sequence number. 6. When all expected packets are present, rebuild chunk. 7. Append chunk to file. 8. Send ACK. 9. Repeat until file_size bytes have been reconstructed.

ACK Rule

Never send the next chunk until the previous chunk has been acknowledged. If an ACK does not arrive after timeout, resend the same chunk. The receiver should only acknowledge a chunk after all packets for that chunk have been received.

Scapy vs Raw Sockets

Use Scapy AsyncSniffer for: - Port knocking - Connect/disconnect requests - File transfer request packets - Run program request packets Use raw sockets for: - File chunks - Chunk ACK packets - Command streams - Program output streams - Program output ACK packets